

CADE BALLEW

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cade-b.github.io

EDUCATION

University of Washington

- Ph.D. in Applied Mathematics 2021–2026
 - Advisor: Thomas Trogdon.
- M.S. in Applied Mathematics 2021–2022

Rice University

- B.A., *magna cum laude* 2017–2021
 - Majors: Computational and Applied Mathematics; Mathematical Economic Analysis.
 - Minor: Mathematics.

PUBLICATIONS AND PREPRINTS

Publications

3. C. Ballew, D. Bilman, and T. Trogdon. Efficient computation of soliton gas primitive potentials. *Journal of Nonlinear Waves*, 1:e12, 2025.
2. C. Ballew and T. Trogdon. The Akhiezer iteration. In *Contemporary Mathematics*, volume 822, pages 1–34. American Mathematical Society, 2025.
1. C. Ballew and T. Trogdon. A Riemann–Hilbert approach to computing the inverse spectral map for measures supported on disjoint intervals. *Studies in Applied Mathematics*, 152(1):31–72, 2024.

Preprints

1. C. Ballew, T. Trogdon, and H. Wilber. The Akhiezer iteration and inverse-free solvers for Sylvester matrix equations. *arXiv preprint 2503.17496*, 2025.

Software

2. C. Ballew, D. Bilman, and T. Trogdon. <https://github.com/cade-b/KdVSolitonGas.jl>, 2025.
1. C. Ballew and T. Trogdon. <https://github.com/cade-b/RecurrenceCoefficients.jl>, 2023.

INVITED TALKS

15. *Numerical methods for problems on cut domains*. Mathematics Colloquium, Baylor University, February 2026.
14. *Numerical methods for KdV soliton gasses*. AMS Fall Southeastern Sectional Meeting, Tulane University, October 2025.
13. *The Akhiezer iteration for matrix functions and Sylvester equations*. Joint SIAM/CAIMS Annual Meetings, Montréal, QC, July 2025.

12. *The Akhiezer iteration for matrix functions and Sylvester equations*. 30th Biennial Numerical Analysis Conference, University of Strathclyde, June 2025.
11. *The Akhiezer iteration for matrix functions and Sylvester equations*. Householder Symposium XXII, Cornell University, June 2025.
10. *Computing KdV soliton gas potentials*. The Thirteenth International Conference on Nonlinear Evolution Equations and Wave Phenomena: Computation and Theory, Athens, GA, April 2025.
9. *Polynomials: Better than you think*. Workshop on Challenges, Opportunities, and New Horizons in Rational Approximation, Banff International Research Station, April 2025.
8. *Computing KdV soliton gas potentials*. AMS Spring Central Sectional Meeting, University of Kansas, March 2025.
7. *The Akhiezer iteration for matrix functions and Sylvester equations*. SIAM Conference on Computational Science and Engineering, Fort Worth, TX, March 2025.
6. *Some numerical applications of Riemann–Hilbert problems*. Joint Mathematics Meetings, Seattle, WA, January 2025.
5. *Numerical solutions of Riemann–Hilbert problems on disjoint intervals*. Integrable Systems and Random Matrix Theory Seminar, University of Michigan, October 2024.
4. *Orthogonal polynomials and Geronimus’s theorem*. Arbeitsgemeinschaft on Quantum Signal Processing and Nonlinear Fourier Analysis, Oberwolfach Research Institute for Mathematics, October 2024.
3. *Applications of numerical solutions of Riemann–Hilbert problems on disjoint intervals*. SIAM Conference on Nonlinear Waves and Coherent Structures, Baltimore, MD, June 2024.
2. *Numerical solutions of Riemann–Hilbert problems on disjoint intervals*. CMS Summer Meeting, University of Saskatchewan, June 2024.
1. *Computing with orthogonal polynomials on disconnected domains*. SIAM PNW Biennial Meeting, Western Washington University, October 2023.

POSTERS

2. *Computing with orthogonal polynomials for integrable systems: A Riemann–Hilbert approach*. SIAM Conference on Nonlinear Waves and Coherent Structures, Baltimore, MD, June 2024.
1. *Computing with orthogonal polynomials on disconnected domains: A Riemann–Hilbert approach*. Workshop on complex analysis: techniques, applications and computations, Isaac Newton Institute, July 2023.

CONFERENCE ORGANIZATION

5. Session co-organizer, SIAM Annual Meeting, Minisymposium on “Orthogonal Polynomials and Special Functions in Mathematical Physics”, Cleveland, OH, July 2026.
4. Session co-organizer, 27th Conference of the International Linear Algebra Society, Minisymposium on “Polynomials, Krylov Methods and Applications”, Blacksburg VA, May 2026.
3. Session co-organizer, Joint SIAM/CAIMS Annual Meetings, Minisymposium on “Numerical methods in the theory of orthogonal polynomials and special functions”, Montréal, QC, July 2025.
2. Session co-organizer, Joint Mathematics Meetings, AMS Special Session on “Recent Advancements in Integrable Systems and Orthogonal Polynomials”, Seattle, WA, January 2025.
1. Session co-organizer, SIAM PNW Biennial Meeting, Session on “Scientific Computing and Numerical Analysis”, Western Washington University, Bellingham, WA, October 2023.

TEACHING EXPERIENCE

University of Washington

- Instructor of Record, AMATH 353 (Partial Differential Equations and Waves), Summer 2024.
- Teaching Assistant, AMATH 352 (Applied Linear Algebra and Numerical Analysis), Winter 2026.
- Teaching Assistant, AMATH 401 (Vector Calculus and Complex Variables), Autumn 2025.
- Teaching Assistant, AMATH 502 (Introduction to Dynamical Systems and Chaos), Winter 2025.
- Teaching Assistant, AMATH 567 (Applied Complex Analysis), Autumn 2023, Autumn 2024.
- Teaching Assistant, CFRM 507 (Optimization Methods in Finance), Autumn 2021, Autumn 2022.

Rice University

- Grader, CAAM 336 (Differential Equations in Science and Engineering), Spring 2019.

SERVICE

- SIAM UW Student Chapter
 - Vice President 2022–2023
 - Outreach Coordinator 2023–2024
- Numerical Analysis Research Club
 - Student organizer Spring 2023, Autumn 2023

AWARDS

- Kevorkian Fellowship 2025
- Wan Fellowship 2021–2024
- AMS Sectional Travel Grant 2025
- Boeing Research Award 2025
- SIAM CSE Student Travel Award 2025
- US Junior Oberwolfach Fellows NSF grant 2024
- SIAM NWCS Student Travel Award 2024
- CMS Student Travel Award 2024
- SIAM PNW Student Travel Award 2023
- Phi Beta Kappa 2021
- Peter Mieszkowski Prize for Honors Program Research 2021
- Malcolm Gillis Award in Mathematical Economic Analysis 2021
- Honors in Economics 2021
- Louis J. Walsh Scholarship 2020–2021
- Michael D. Maher RISE Award in Economics 2020
- Rice University President's Honor Roll (5 semesters)